Rotaract Club of Xiamen University Malaysia

Rotary Club of Kuala Lumpur Dynamics

Rotary Club of Tun Razak Exchange

SDC Hackathon

Technical Tools Declaration

Hosted by Xiamen University Malaysia (XMUM) AI Club

Submission Deadline: [Insert date]

Instructions: Declare every API, cloud service, AI tool, dataset, model, and sponsor resource your team used to build the project. Complete every section; where a category does not apply, tick "No" or write "None" – do not leave it blank. Submit as a single PDF document named "SDC2026_TeamName_TechDeclaration.pdf" together with your project submission. Undeclared third-party resources discovered at judging may affect your score or eligibility.

Instagram: @sdc_xmumhack ***Email:*** sdc.xmumhack@gmail.com

1. TEAM AND PROJECT OVERVIEW

Team Name

4bit

Project / Solution Name

Skilytics

Name (Leader)

Muhammad Ameer Yusoff Bin Azeri

Email & Phone (Leader)

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University Affiliation(s)

Universiti Teknologi Malaysia

Live Prototype URL

https://skilytics.vercel.app

Code Repository URL (public / access granted to judges)

https://github.com/ameer-yusoff/SDC2026_Hackathon_4bit_Skilytics

2. ARCHITECTURE & TECHNOLOGY STACK

2.1. System Architecture Overview

Describe how the components declared in this document connect end-to-end (e.g., user interface → backend → AI model/API → database → cloud hosting). Attach a system block diagram as an appendix if available.

[Maximum 200 words]

- User Interface (Frontend): A React single-page application built with Vite (src/artifacts/reskill-ai) that captures user interactions, profile configurations, and RIASEC assessment responses.
- Backend API: A Node.js Express server that processes client REST requests, handling routing, authentication, and core evaluation logic.
- AI Model / API: Interacts with AI agent services (src/artifacts/api-server/src/lib/llm.ts) to process CVs, evaluate skill gaps, and generate AI-driven career recommendations.
- Database Layer: Managed via Drizzle ORM (src/lib/db) connected to a Neon Serverless PostgreSQL database, storing user profiles, skill matrices, and course catalog items.
- Cloud Hosting: Cloud infrastructure deployed via Render (render.yaml) for backend deployment and Vercel (vercel.json) for frontend hosting.

2. ARCHITECTURE & TECHNOLOGY STACK

2.2. Languages, Frameworks & Libraries

List the main programming languages, frameworks, and third-party libraries that your team did not create. Open-source use is permitted provided it is declared and properly licensed/attributed.

Name & Version	Purpose in Project	License
TypeScript (v5.x)	Primary programming language used across both frontend and backend for type-safe application development.	Apache 2.0
Node.js (v20.x)	Runtime environment powering the backend API server, automated build scripts, and local developer environment.	MIT
Python (v3.10+)	Scripting language used for model testing, sending structured inference requests to RunPod, and data validation.	PSF License
React (v18.x)	Frontend library for building the interactive user interface, assessment workflows, and dashboard components.	MIT
Vite (v5.x)	Frontend tooling and development server providing fast bundling and hot-module replacement.	MIT
Tailwind CSS (v3.x)	Utility-first CSS framework used for styling responsive, modern user interface layouts.	MIT

Name & Version	Purpose in Project	License
Radix UI / Shadcn UI (Latest)	Unstyled, accessible component primitives used to construct dialogs, tabs, progress bars, and forms.	MIT
Lucide React (v0.x)	Icon library providing consistent vector icons across the platform interface.	ISC
TanStack React Query (v5.x)	Asynchronous state management library handling API data caching, background fetching, and server state synchronization.	MIT
Express.js (v4.x)	Minimalist backend web framework routing API requests for CV processing, user profiling, and market data queries.	MIT
Zod (v3.x)	Schema declaration and validation library enforcing structured, type-safe inputs and outputs for API routes and AI payloads.	MIT
Orval (v7.x)	OpenAPI code generator creating type-safe TypeScript React query hooks and fetch clients directly from OpenAPI specs.	MIT
Drizzle ORM (v0.x)	TypeScript Object-Relational Mapper used for database schema declaration, migrations, and type-safe database queries.	Apache 2.0
Neon Serverless SDK (@neondatabase/serverless)	Serverless PostgreSQL driver connecting the backend to cloud Neon database instances via WebSockets/HTTP.	Apache 2.0 / MIT
Google Generative AI SDK (@google/genai)	Client SDK for interacting with the Gemini API to execute resume parsing, interest extraction, and career reasoning.	Apache 2.0

Name & Version	Purpose in Project	License
Tesseract.js (v7.x)	Primary OCR engine to extract text from image/scanned CV files (scanned images like JPEG/PNG).	Apache 2.0
PDF.js (pdfjs-dist & unpdf) (v4.x / v1.x)	Extracts text, structure, and metadata from PDF-formatted CV files.	Apache 2.0
Recharts (v2.x)	Generates interactive data visualizations, radar charts (6 competency scores), and analytics charts in the dashboard	MIT
Framer Motion (v12.x)	Handles micro-animations, score card transitions, and smooth UI interactivity.	MIT
Wouter (v3.x)	Client-side routing library used for frontend page navigation (not React Router).	MIT
node-postgres (pg) (v8.x)	The actual driver (driver) that connects the Node.js backend directly to the Neon Postgres database.	MIT
Pino & Pino-HTTP (v9.x)	High-performance logging system for backend error monitoring and request auditing.	MIT

3. APIs & CLOUD SERVICES

3.1. External APIs

List every external API your solution calls (GenAI, maps, payments, messaging, data feeds, etc.). Do not include API keys or secrets in this document.

API and Provider	Purpose in Project	Access Tier (free / paid / sponsor-provided)
Google Gemini API	GenAI reasoning engine for CV parsing, RIASEC score calculation, skill gap analysis, and personalized roadmap generation.	Free
RunPod Serverless Inference API	REST API endpoint used by the backend to trigger serverless inference tasks on custom open-source models (Qwen).	Sponsor-provided
OpenDOSM API	Live national labor force statistics feed providing monthly employment data and local job market benchmarks.	Free
Anthropic Economic Index Feed	Research data feed providing country-level AI usage metrics and task-level AI exposure ratings by SOC occupation code.	Free
O*NET Web Services API	Standardized occupational database API providing official job taxonomy, skill definitions, and RIASEC job mappings.	Free
Microsoft Learn Catalog API	Ingests verified public course modules, duration estimates, and certification pathways.	Free
freeCodeCamp Content API	Curriculum feed supplying structured software development learning tracks and free certification units.	Free

Name & Version	Purpose in Project	Access Tier (free / paid / sponsor-provided)
GitHub REST API	Content API retrieving practical coding exercises, repository templates, and technical hands-on tasks.	Free
Hugging Face Hub API	Content and model registry API providing machine learning course modules and open tutorial resources.	Free

3. APIs & CLOUD SERVICES

3.2. Cloud Platforms & Deployment

List every cloud service used for hosting, storage, databases, authentication, or compute, and indicate how the prototype is deployed.

Service & Provider	Purpose in Project	Plan / Region
RunPod	Serverless cloud GPU compute hosting Docker containers for open-source decision models (Qwen) with automatic scale-to-zero.	Sponsor-provided
Neon	Serverless PostgreSQL cloud database managing user profiles, assessment submissions, cohort statistics, and market cache.	Free
Render	Cloud application platform hosting and running the backend Express API server container (render.yaml).	Free
Vercel	Edge hosting platform providing continuous deployment (CI/CD) and global CDN for the React/Vite frontend (vercel.json).	Free
Replit	Cloud IDE and staging environment used for interactive prototyping, testing, and multi-agent development (.replit).	Sponsor-provided
GitHub	Cloud Git repository hosting, version control, source code management, and automated webhooks.	Free

4. AI TOOLS, MODELS & DATASETS

4.1. AI Models Used in the Solution

List every AI model embedded in or called by your application, including pre-trained, fine-tuned, and self-trained models.

Model & Version	Source / Provider	Usage (pre-trained / fine-tuned / self-trained)	License / Terms
Gemini-flash-lite 3.1	Google AI Studio	Pre-trained	Google APIs Terms of Service
Qwen 3.5-27b Qwen3.5-9b	Alibaba Cloud	Pre-trained	Apache 2.0 / Qwen Open License

4. AI TOOLS, MODELS & DATASETS

4.2. Datasets

List all datasets used for training, fine-tuning, evaluation, or in the running application. State clearly if a dataset contains personal or sensitive data.

Dataset Name	Source / URL	Type (public / private / self-collected / synthetic)	License	Personal Data? (Y/N)
OpenDOSM Monthly Labour Force	https://data.gov.my/data-catalogue/lfs_month	Public	CC BY 4.0	N
AI Usage by Country (Anthropic Economic Index)	https://huggingface.co/datasets/Anthropic/EconomicIndex	Public	CC BY 4.0	N
AI Exposure by Occupation (Anthropic Economic Index)	https://huggingface.co/datasets/Anthropic/EconomicIndex/tree/main/labor_market_impacts	Public	CC BY 4.0	N
O*NET Occupation References (v31.0)	https://www.onetcenter.org/database.html	Public	CC BY 4.0	N
Microsoft Learn Public Course Catalogue	https://learn.microsoft.com/api/catalog/?locale=en-us&type=learningPaths	Public	Microsoft Terms of Use	N
freeCodeCamp Public Curriculum	https://github.com/freeCodeCamp/freeCodeCamp/blob/main/curriculum/structure/curriculum.json	Public	CC BY-SA 4.0 / BSD 3-Clause	N
GitHub Skills Exercises	https://api.github.com/orgs/skills/repos?per_page=100&type=public	Public	MIT / GitHub Terms	N
Hugging Face Learn Catalogue	https://huggingface.co/learn	Public	Open Access / CC BY 4.0	N

Dataset Name	Source / URL	Type (public / private / self-collected / synthetic)	License	Personal Data? (Y/N)
Holland Code (RIASEC) Question Set	https://www.onetcenter.org/dl_files/Mini-IP.pdf	Public	Public Domain	N
HubSpot Academy Public Catalogue	https://academy.hubspot.com/courses	Public	HubSpot Terms of Service	N
DOSM JMI Q4 2025 · Selected Occupation Snapshot	https://blog.gdeltproject.org/gdelt-doc-2-0-api-debuts/	Public	CC BY 4.0 / Open Data	N
Google News Public Career RSS	https://news.google.com/rss/search?q=%28upskilling+OR+reskilling+OR+%22workforce+skills%22+OR+%22AI+workplace%22+OR+%22hiring+demand%22%29+Malaysia+when%3A30d&hl=en-MY&gl=MY&ceid=MY%3Aen	Public	Google Terms of Service	N
BLS US Career Growth Reference (2025–2035)	https://www.bls.gov/emp/tables/occupational-projections-and-characteristics.htm	Public	Public Domain (U.S. Govt)	N
O*NET Content Model · Skill References	https://www.onetcenter.org/content.html	Public	CC BY 4.0	N
United Nations · SDG Framework	https://sdgs.un.org/goals	Public	UN Open License / CC BY 3.0 IGO	N
OpenLearn · Reviewed Public Courses	https://www.open.edu/openlearn/	Public	CC BY-NC-SA 4.0 / Open University Terms	N

Dataset Name	Source / URL	Type (public / private / self-collected / synthetic)	License	Personal Data? (Y/N)
Himalayas · Public Vacancies	https://www.onetcenter.org/dl_files/Mini-IP.pdf	Public	Himalayas Terms of Use / Keyless AP	N
Remotive · Public Remote Vacancies	https://academy.hubspot.com/courses	Public	Remotive API Terms / Keyless API	N
RemoteOK · Public Job Feed	https://blog.gdeltproject.org/gdelt-doc-2-0-api-debuts/	Public	RemoteOK Public Terms / Keyless API	N
Synethetic Data Graduate Employment	https://github.com/ameer-yusoff/SDC2026_Hackathon_4bit_Skilytics/blob/main/synthetic%20data/graduate-employment-2024-2026.json	Synthetic	CC0 1.0	N
Career Personas	https://github.com/ameer-yusoff/SDC2026_Hackathon_4bit_Skilytics/blob/main/synthetic%20data/career-personas.json	Synthetic	CC0 1.0	N
Demo CV #1-3	https://github.com/ameer-yusoff/SDC2026_Hackathon_4bit_Skilytics/tree/main/synthetic%20data	Synthetic	CC0 1.0	N

4. AI TOOLS, MODELS & DATASETS

4.3. AI-Assisted Development Tools

Tools used to help write code, design, content, or documentation (e.g., chat assistants, code completion tools, image generators). Declaring these tools does not reduce your score; undeclared use may.

Did your team use any generative-AI tool to help create code, design, text, or other project material?

☑ Yes ☐ No

If yes, list them below:

Tool & Version	Used For	Extent of Use (minor / moderate / major)
Antigravity	Core agentic code generation, architecture planning, and debugging multi-agent workflow pipelines.	Major
OpenAI Codex	TypeScript/React code completion, helper function drafting, schema-bound logic, and unit test generation.	Moderate
Replit	Full-stack project scaffolding, rapid UI mockup prototyping, sandbox testing, and server environment setup.	Minor
Google Gemini	Prototyping multi-agent system prompts, designing structured Zod schemas, generating multi-language dictionaries (EN/BM/ZH), and technical documentation.	Major

4. AI TOOLS, MODELS & DATASETS

Briefly describe how your team reviewed, tested, and took responsibility for AI-generated output:

[Maximum 100 words]

We treated all AI-generated code and model responses as rough drafts that required human checks. Our team manually reviewed and tested every piece of code before merging it into GitHub.

For AI model outputs, we enforced strict Zod schema validation to stop errors and hallucinations before data reached the frontend. We also ran automated unit tests using mock CV profiles to check parsing accuracy and course recommendations. Ultimately, our team took full ownership by testing the pipeline end-to-end to ensure all career guidance was accurate, safe, and reliable.

5. SPONSOR-PROVIDED RESOURCES

Declare any official SDC sponsor-provided GenAI API, cloud credits, tools, mentoring, or data that your team used.

Did your project use any official SDC sponsor-provided resource?

☑ Yes ☐ No

If yes, list them below:

Sponsor	Resource (API / credits / platform / data)	How It Was Used in the Project
Replit	Cloud IDE platform, development workspace, and Replit Agent tools	Used to set up the project workspace, build and test full-stack components, and prototype interactive UI flows.
RunPod	Cloud GPU credits and serverless container endpoints (H100 GPUs).	Used to test open-source decision models (Qwen), configure scale-to-zero serverless workers, and run structured inference via Python and cURL.

5. SPONSOR-PROVIDED RESOURCES

Was the sponsor resource essential to the core functionality of the prototype, or could it be replaced? Explain briefly:

[Maximum 100 words]

The sponsor resources were very helpful for fast prototyping, but they could be replaced if necessary.

RunPod allowed us to test open-source AI models on cloud GPUs without needing expensive local hardware, and Replit made it easy for our team to build and test our code together in a shared cloud workspace.

However, our core features do not depend exclusively on them. Because our platform uses standard web frameworks and flexible containerized models, RunPod could be replaced by other cloud GPU providers or commercial AI APIs, and Replit could be replaced by regular tools like VS Code.

6. SECURITY, PRIVACY & RESPONSIBLE USE

6.1. Data Handling & Privacy

What user or personal data does your application collect, store, or send to third-party services (including AI APIs)? How is it protected, and how long is it kept? State "No user data is collected" if applicable.

[Maximum 150 words]

Skilytics collects basic account information (name and email), uploaded resumes (education, work experience, and skills), and answers from the career interest quiz. Resume text and quiz answers are sent to AI services (Google Gemini API and RunPod) strictly to extract skills, evaluate career paths, and create personalized reskilling roadmaps. We never sell or share user data with advertisers.

All data is stored in a secure cloud database (Neon Postgres) using encrypted connections (HTTPS/SSL). System credentials and API keys remain protected on our backend server so sensitive data is never exposed to the public, and our AI service providers do not use submitted data to train their public models.

Personal data is kept only while the user's account is active to display their career progress. Users can update or remove their profile at any time, which permanently removes their stored records.

6. SECURITY, PRIVACY & RESPONSIBLE USE

6.2. Known Limitations & Risks

Identify the main technical limitations of your AI components (e.g., hallucination, bias, rate limits, dependency on a single provider) and any mitigation you have implemented.

[Maximum 150 words]

- CV Parsing Inaccuracies: AI models can misinterpret complex resume formatting or extract skills not actually present in a CV.
- Mitigation: We give users an interactive review screen to verify and edit parsed skills before processing, and we validate extracted terms against verified O*NET skill lists.

- API Downtime & Rate Limits: External services can time out.
- Mitigation: We cache frequent queries and store fallback data snapshots.

- AI Hallucinations: AI can invent courses or career facts. Mitigation: We use Zod schemas to block malformed outputs and only recommend verified courses from our connected catalogues.

- Provider Dependency: Relying on one vendor risks system failure.
- Mitigation: Our architecture supports switching between open-source models (Qwen) on RunPod and Google Gemini

7. COMPLIANCE & READINESS

Tick all boxes that apply before submitting.

- ☑ All APIs, cloud services, AI tools, models, and datasets used in the project are declared in this document.
- ☑ All third-party code, datasets, and models are used in accordance with their licenses, and attribution is provided where required.
- ☑ No API keys, passwords, or other credentials are exposed in the code repository, prototype, or this document.
- ☑ Any use of generative-AI tools for development has been disclosed and reviewed by the team.
- ☑ Any use of official SDC sponsor resources has been declared in Section 5.
- ☑ The prototype described here is accessible to judges through the links provided in Section 1.
- ☑ All fields in this template have been completed within the word limits stated.

8. DECLARATION

By submitting this document, the team leader confirms that the information provided is complete and accurate to the best of the team's knowledge, and that every API, cloud service, AI tool, dataset, model, and sponsor resource used in the project has been declared.

[Signature]



[Date]

24/10/2026